

Cells: Structure, Transport & Division

Test · 33 marks total

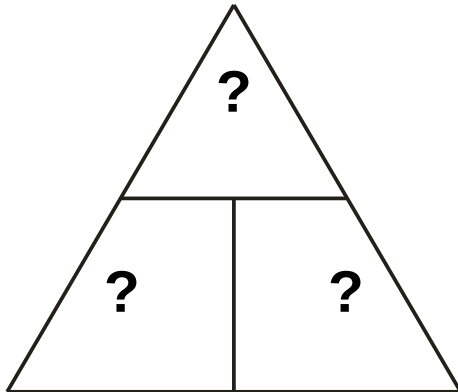
Name: _____

Date: _____

Score: ____ / 33

Section A — Complete the formula triangle

A1. Fill in the three blanks with I, M and R in the correct places.



[3 marks]

Section B — Straight substitution

B1. A cell has an image size of 6 mm and a real size of 0.3 mm. Calculate the magnification.

[2 marks]

B2. A structure has an image size of 15 mm and a real size of 0.05 mm. Calculate the magnification.

[2 marks]

Section C — Rearrange and solve

C1. A real structure is 0.002 mm across and is magnified $\times 500$. Calculate the image size, in mm. Show how you rearranged the equation.

[3 marks]

C2. An image measures 4 mm across at a magnification of $\times 800$. Calculate the real size, in mm. Show how you rearranged the equation.

[3 marks]

C3. A real structure is 20 μm across and is magnified $\times 50$. Calculate the image size, in mm. (Convert units first.)

[3 marks]

Section D — Cell structures

D1. Name two structures found in a plant cell but not in an animal cell.

[2 marks]

D2. Name two structures found in a bacterial cell that are not found in an animal cell.

[2 marks]

D3. State the function of the mitochondria.

[1 mark]

D4. State the function of a chloroplast.

[1 mark]

Section E — Transport across membranes

E1. Define diffusion.

[2 marks]

E2. Define osmosis, and state what makes it different from diffusion of a dissolved substance.

[2 marks]

E3. Explain why active transport requires energy from respiration, using the example of mineral ion uptake by plant root hairs.

[2 marks]

Section F — Cell division & differentiation

F1. State two reasons why body cells divide by mitosis.

[2 marks]

F2. State the number of chromosomes in a normal human body cell and in a human gamete, and explain why the gamete has this number.

[3 marks]